MATH 11 — ELEMENTARY STATISTICS

§7.2

The Central Limit Theorem

The shape of the distribution of sample means

Today's Goal

After completing this section, you should be able to:

- State the Central Limit Theorem
- Determine when the Central Limit Theorem can be applied
- Calculate probabilities for sample means using the Central Limit Theorem
- Find values from known probabilities for sample means

Learning Target 12

LT.12 The Central Limit Theorem

I understand when the Central Limit Theorem can be used and what it tells us regarding the sampling distribution of sample means.

Bags of Coffee

A machine fills bags of coffee. The weights of the bags are normally distributed with a mean of 340 grams and a standard deviation of 6 grams.

Which is more likely to be less than 338 grams: the weight of one bag, or the mean weight of 16 bags?

The Central Limit Theorem

Theorem 7.2.1 The Central Limit Theorem

For samples of size $n > 30$, the sampling distribution of $\bar{x}$ can be approximated by a normal distribution with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

In Symbols

In symbols, we write this as

$$\bar{X} \sim N\left(\mu_{\bar{x}}, \sigma_{\bar{x}}\right) = N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

By the Central Limit Theorem, as n grows, the distribution of sample means more closely follows a normal distribution. Note that this happens regardless of whether the population is normally distributed.

When the population is normally distributed, the Central Limit Theorem applies for any sample size n.

Converting to z-scores

When applying the Central Limit Theorem, we convert sample means to z-scores using the formula:

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}.$$

Using Excel for CLT Probabilities

Procedure 7.2.2 Using Excel for CLT Probabilities

When working with the Central Limit Theorem, use `NORM.DIST` with the sampling distribution parameters:

```
=NORM.DIST(x_bar, mean, standard_dev/SQRT(n), TRUE)
```

- `x_bar` = the sample mean value
- `mean` = the population mean μ
- `standard_dev/SQRT(n)` = the standard error $\frac{\sigma}{\sqrt{n}}$
- `TRUE` = cumulative probability (always use `TRUE`)

Example: For $\mu = 172, \sigma = 10, n = 20$, to find $P(\bar{x} < 175)$:

```
=NORM.DIST(175, 172, 10/SQRT(20), TRUE)
```

One Bag

A machine fills bags of coffee. The weights of the bags are normally distributed with a mean of 340 grams and a standard deviation of 6 grams.

(a) Find the probability that one randomly selected bag weighs less than 338 grams.

(b) Find the probability that a random sample of 16 bags has a mean weight of less than 338 grams.

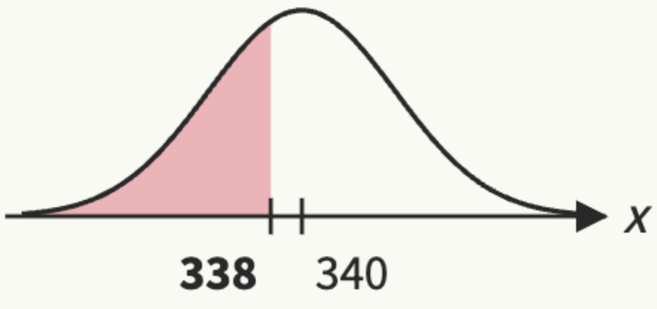
One Bag

(a) Note that this does not require the use of the Central Limit Theorem. We are only selecting one individual from the population, which is normally distributed with $\mu = 340$ and $\sigma = 6$. So we can just apply what we already learned from Chapter 6.

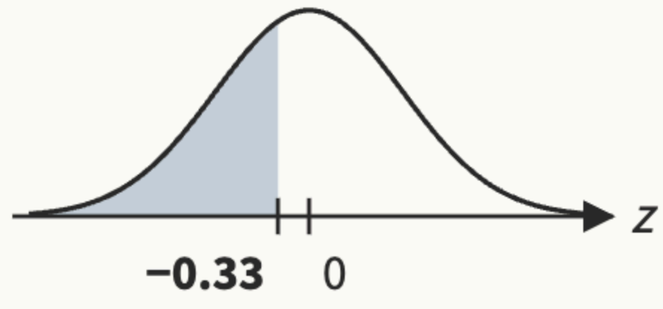
We are being asked to find $P(x < 338)$.

One Bag

(a) Converting to the standard normal distribution:



$$z = \frac{338 - 340}{6} \approx -0.33$$



Using Table A-1, we get $P(x < 338) = P(z < -0.33) = 0.3707$.

The Mean of 16 Bags

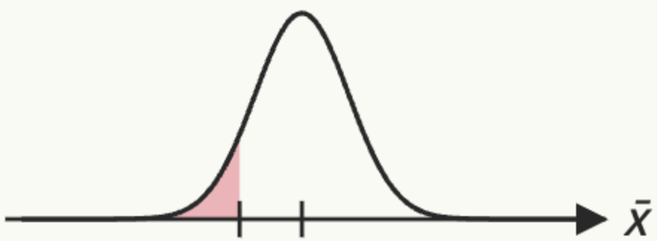
(b) This time, we are selecting a sample of 16 bags and are being asked to find $P(\bar{x} < 338)$. This means that we need to use the Central Limit Theorem.

Thankfully, the population is normally distributed, so we may use the Central Limit Theorem even though the sample size is less than 30.

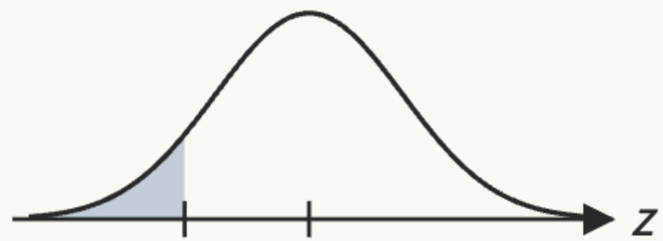
The Central Limit Theorem tells us that $\bar{x} \sim N\left(340, \frac{6}{\sqrt{16}}\right) = N(340, 1.5)$.

The Mean of 16 Bags

(b) Converting to the standard normal distribution:



$$z = \frac{338 - 340}{\frac{6}{\sqrt{16}}} \approx -1.33$$



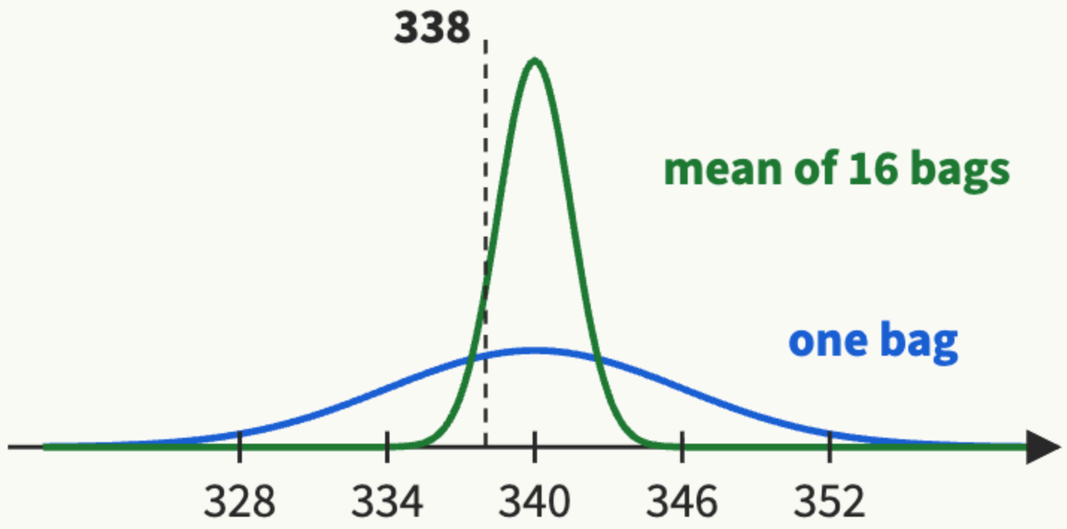
Using Table A-1, we get $P(\bar{x} < 338) \approx P(z < -1.33) = 0.0918$.

Excel: `=NORM.DIST(338,340,6/SQRT(16),TRUE)` returns the more precise value 0.0912.

Comparing the Two Distributions

Notice that the distribution of bag weights and the distribution of sample means of bag weights are both normally distributed with the same mean.

However, the distribution of sample means has a lower standard deviation. This tells us that the distribution of bag weights is more spread out than the distribution of sample means, which is why it makes sense that $P(x < 338) > P(\bar{x} < 338)$.



Less Than How Many Grams?

As in Example 5, the weights of the bags are normally distributed with a mean of 340 grams and a standard deviation of 6 grams.

If a random sample of 16 bags is selected, then 10% of the time, their mean weight will be less than how many grams?

We are looking for a value of $\bar{x} = a$ for which $P(\bar{x} < a) = 0.10$. The sample means have a mean of 340 grams and standard deviation of $\frac{6}{\sqrt{16}} = 1.5$ grams. We will find a_z for which $P(z < a_z) = 0.10$ and then convert back to a.

Less Than How Many Grams?

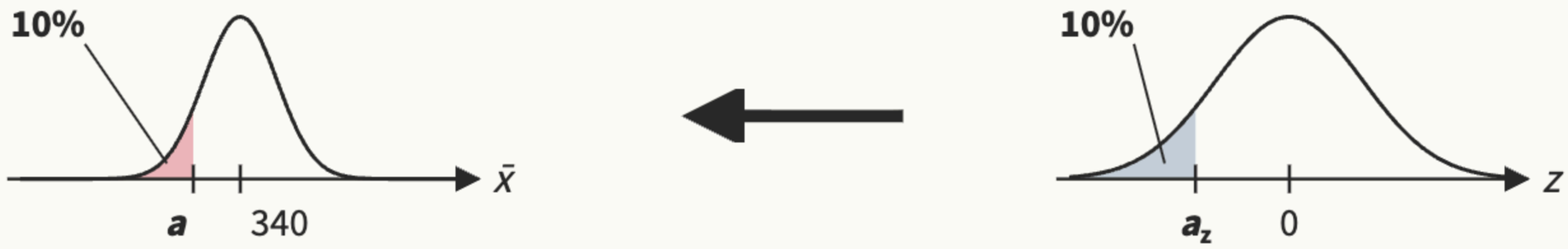


Table A-1 gives us $a_z \approx -1.28$, since the closest area to 0.10 is 0.1003.

Converting back to a gives us

$$\begin{aligned} a &= \mu_{\bar{x}} + a_z \cdot \sigma_{\bar{x}} \\ &\approx 340 + (-1.28)(1.5) = 338.08. \end{aligned}$$

So the sample mean will be less than about 338.08 grams 10% of the time. Excel: `=NORM.INV(0.10,340,6/SQRT(16))` returns 338.0777.

Commute Times

Suppose the commute times of students at a college are skewed right, with a mean of 24 minutes and a standard deviation of 8 minutes. A random sample of 40 students is selected.

(a) Can we use the Central Limit Theorem for the sample mean? Why?

Yes. $n = 40 > 30$, so it applies even though the population is skewed.

(b) Find the standard error $\sigma_{\bar{x}}$.

$\frac{8}{\sqrt{40}} \approx 1.2649$ **minutes**

(c) Find $P(\bar{x} > 26)$.

$z \approx 1.58$**, so** $1 - 0.9429 = 0.0571$ **(Excel** 0.0569**)**

Commute Times

Suppose the commute times of students at a college are skewed right, with a mean of 24 minutes and a standard deviation of 8 minutes. A random sample of 40 students is selected.

(d) 90% of the time, the sample mean is less than how many minutes?

$24 + (1.28)(1.2649) \approx 25.62$ **minutes (Excel** 25.62**)**

(e) Can we find the probability that one student's commute is more than 26 minutes the same way?

No. One student comes from the population, which is skewed, not normal.